Numerical Design and Performance Mapping of a Micro Cold-Gas Thruster for Small Satellites

1 Introduction

Converging–diverging (C–D) nozzles are widely used in rocket engines and micro-thrusters to accelerate gas to supersonic speeds [1, 2]. In this project, a numerical model is developed to simulate compressible flow through a variable-area nozzle using a quasi-one-dimensional formulation. The solver computes the distribution of Mach number, pressure and velocity along the nozzle and is validated using analytical relations and CFD simulations.

2 Literature Review

Numerical simulation of compressible nozzle flows is commonly performed using quasi-one-dimensional models in which flow properties are assumed uniform across each cross-section while the nozzle area varies axially [3]. Despite this simplification, such models capture the essential physics of compressible flow acceleration and choking in converging–diverging ducts.

Putra [4] solved the quasi-one-dimensional Euler equations using numerical schemes such as the MacCormack predictor–corrector method. Finite-volume formulations are often preferred because they preserve conservation laws across control volumes and provide improved robustness for hyperbolic systems [5].

Flux-based schemes are used to compute interface fluxes of mass, momentum and energy. Among these, the Lax–Friedrichs method is frequently used in preliminary studies due to its simplicity and numerical stability. Explicit Runge–Kutta schemes are typically employed for time integration, with the time step determined using the Courant–Friedrichs–Lewy (CFL) stability condition [6].

3 Quasi- 1D Numerical Modeling

A common approach in the literature is the quasi-one-dimensional nozzle model, where flow properties are assumed uniform across each cross-section while the nozzle area varies along the axial direction. This simplifies the governing equations while still capturing the key physics of compressible flow acceleration.

The governing Euler equations for compressible flow are:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{V}) = 0 \tag{1}$$

$$\frac{\partial (\rho \mathbf{V})}{\partial t} + \nabla \cdot (\rho \mathbf{V}\mathbf{V}) + \nabla p = 0 \tag{2}$$

$$\frac{\partial E}{\partial t} + \nabla \cdot [(E + p)\mathbf{V}] = 0 \tag{3}$$

For a variable-area nozzle these equations can be written in conservative form as

$$\frac{\partial U}{\partial t} + \frac{\partial F}{\partial x} = J \tag{4}$$

where U represents the conserved variables, F the flux terms and J the geometric source term resulting from the change in nozzle area.

4 Numerical Solver Framework

A quasi-one-dimensional solver is implemented to compute the axial distribution of Mach number, pressure and velocity. The nozzle geometry is represented using a smooth area function forming a converging section, throat and diverging section.

The governing equations are solved using the finite volume method (FVM) [5]. The domain is discretized into computational cells and conservation equations are applied within each control volume. Interface fluxes are computed using the Lax–Friedrichs scheme, which introduces controlled numerical dissipation and improves stability when solving hyperbolic systems.

Time integration is performed using an explicit Runge–Kutta method. The time step is determined from the CFL condition

$$\Delta t = \mathrm{CFL}\frac{\Delta x}{|u| + a} \tag{5}$$

where u is the flow velocity and a is the local speed of sound. The solution is advanced iteratively until convergence, yielding the axial distributions of Mach number, pressure and velocity.

5 Preliminary Results

Preliminary simulations were carried out for the converging section of an axisymmetric nozzle assuming steady, one–dimensional compressible flow with choking at the throat ($M = 1$). The inlet conditions used were $p_i = 2.85$ bar, $T_i = 600$ K, inlet diameter $D_i = 810$ mm, expansion ratio $A/A^* = 2.35$, and specific heat ratio $\gamma = 1.4$. Using the isentropic area–Mach relation together with the continuity equation, a nonlinear equation for the Mach number was derived and solved numerically in Python. For the specified area ratio, the analytical isentropic

solution yields two possible Mach numbers, $M = 0.256$ (subsonic) and $M = 2.376$ (supersonic). Since only the converging section was simulated, the subsonic branch is relevant. The numerical solver predicted an inlet Mach number of $M = 0.2727$ and an exit Mach number of $M = 1.0045$, compared with analytical values of $M = 0.256$ and $M = 1.00$, respectively. This corresponds to errors of 6.5% at the inlet and 0.45% at the throat, indicating good agreement with isentropic theory and correct capture of the choking condition. The corresponding simulation results are presented in the plots below:

Analytical Mach number
Simulated Mach number

6 Validation

The numerical solver is validated in two ways. First, results are compared with analytical relations for quasi-one-dimensional compressible nozzle flow. Second, predictions are compared with Computational Fluid Dynamics (CFD) simulations performed in ANSYS Fluent.

6.1 Analytical Validation

For isentropic flow through a converging–diverging nozzle, the Mach number distribution is related to the area ratio by the area–Mach relation [2]

$$\frac{A}{A^*} = \frac{1}{M}\left[\frac{2}{\gamma+1}\left(1+\frac{\gamma-1}{2}M^2\right)\right]^{\frac{\gamma+1}{2(\gamma-1)}} \tag{6}$$

where A^* is the throat area and γ is the specific heat ratio. Once the Mach number distribution is obtained, the pressure and velocity fields are calculated using standard isentropic compressible-flow relations.

6.2 CFD Validation

CFD simulations are performed in ANSYS Fluent [1, 7]. The flow is modeled in a two-dimensional axisymmetric nozzle geometry with air treated as an ideal gas. A density-based compressible solver is used with pressure inlet and pressure outlet boundary conditions.

After convergence, axial distributions of Mach number, pressure and velocity are extracted and compared with the predictions of the quasi-one-dimensional solver. Agreement between these results provides validation of the numerical model.

References

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7 Contribution

Sidhwa Tanuj Haresh (230003073) contributed to the literature review on compressible nozzle flow and participated in the development of the numerical solver. Moving forward, he will focus on further development of the computational framework and generation of the final simulation results.

Tejas Santosh Bhavekar (230003079) contributed to the literature review and worked on the analytical formulation of the problem, including the implementation of isentropic relations and area–Mach calculations. In the next phase of the project, he will perform CFD simulations using ANSYS and verify the numerical results through analytical comparisons.

Siddharth Vezzu (230021016) contributed to the literature review and participated in the implementation of the numerical solver. In the upcoming stages, he will continue working on improving the code implementation and assist with the preparation and documentation of the project results.

Sudarsanam Sai Sri Narasimha (230021018) contributed to the literature review and assisted in preparing the project documentation. In the next phase, he will work on CFD simulations using ANSYS and continue contributing to documentation and report preparation.